Shelf Voice — 08 Deployment and Operations Runbook

Oct 27, 2026 · @Aradhya Chaudhary

Status and open decisions

Version 1.0, approved by the owner on Oct 27, 2026. All six operating decisions recorded below.

#	Decision	Options (recommendation %)	Why the top option	Owner's choice
R1	Package manager	pnpm (65%); npm (35%)	Fast installs, strict dependency resolution, supported natively by Vercel	pnpm
R2	Branching	Trunk-based: short feature branches, PR into main, main is production (65%); main to staging, release branch to production (35%)	Matches Vercel Hobby (production = main); fewer moving parts for 3 people and 8 days	Trunk-based
R3	Production deploys	Automatic on merge to main once CI is green (55%); manually promote a checked preview (45%)	Fast feedback during the build week; switch to manual promotion during the freeze (R4)	Automatic on green merge; manual promotion from the freeze
R4	Code freeze	Features freeze Nov 3, 11:59 PM IST; Nov 4 for fixes, demo video and submission (70%); freeze Nov 4 at noon IST (30%)	A full day to record the video and fix only what it exposes, before the Nov 5, 7:30 AM IST deadline	**Freeze Nov 4 at noon IST** (owner's call over the 70% option); fixes, video and submission in the final ~19 hours

#	Decision	Options (recommendation %)	Why the top option	Owner's choice
R5	Domain	Free Vercel subdomain, e.g. shelfvoice.vercel.app (65%); buy a custom domain (35%)	Zero cost and setup; QR codes and links work immediately	Free Vercel subdomain
R6	Sharing secrets in the team	Each teammate has their own dashboard login; shared values kept in a free password-manager vault (70%); one person holds every secret (30%)	No secrets in chat or the repo; no single point of failure during judging	Own logins + shared vault

Environments, accounts and owners

Three environments, each with its own database and keys, and every account has two teammates with access so nobody is a single point of failure.

Environment	App URL	Database	Meta key	Deploys from
Local	http://localhost:3000	Supabase CLI local stack (Docker)	Personal dev key or recorded responses	Your machine
Preview	Per-PR Vercel preview URL	Staging Supabase project (Mumbai)	Staging key	Every push to a PR branch
Production	shelfvoice.vercel.app (R5)	Production Supabase project (Mumbai)	Production key	main (R2, R3)

Account	Primary	Backup	Notes
GitHub repo (public)	Teammate A	Teammate B	Branch protection on main: PR + green CI required

Account	Primary	Backup	Notes
Vercel project (Hobby)	Teammate A	Teammate B (via the repo; Hobby is one seat)	Production env vars set here
Supabase org (staging + production)	Teammate B	Teammate A	Owner roles for both; two-factor on
Meta Model API team	Teammate A (hackathon account)	Teammate C	Keys per environment; rotate after the hackathon
Google Cloud OAuth client	Teammate B	Teammate A	Consent screen in production mode
Cloudflare (Turnstile), Axiom, Better Stack	Teammate C	Teammate A	Free plans
Discord alert channel	Teammate C	All	Webhook URL stored as a secret

Teammates A, B and C are placeholders; fill in names on day 1.

Local development setup

A new teammate goes from clone to a running app with seeded data in about 15 minutes.

Required software:

Tool	Version	Why
Node.js	24.x (pinned in `package.json` engines and `.nvmrc`)	Vercel's default runtime; still in maintenance LTS until Apr 2028
pnpm	9 or newer (R1)	Package manager
Docker Desktop	Current	Runs the local Supabase stack
Supabase CLI	Current	Local database, migrations, type generation
Git	2.40 or newer	Version control
Chrome	Current	Playwright fake-microphone tests

Steps:

```
git clone https://github.com/<org>/shelf-voice.git && cd shelf-voice
pnpm install
cp .env.example .env.local        # fill values from the team vault (R6)
supabase start                    # local Postgres, Auth, Storage
supabase db reset                 # applies all migrations + seed.sql
pnpm dev                          # http://localhost:3000
```

Everyday commands:

Command	Does
`pnpm dev`	Run the app locally
`pnpm lint` and `pnpm typecheck`	ESLint and TypeScript strict
`pnpm test`	Vitest unit and integration tests
`supabase test db`	pgTAP RLS tests
`pnpm e2e`	Playwright end-to-end with recorded AI
`pnpm eval`	promptfoo evals against the live Meta API (spends credit)
`pnpm record:meta`	Refresh recorded Meta responses after a prompt change
`supabase migration new <name>`	New migration file
`pnpm db:types`	Regenerate database types

Local AI calls use recorded responses by default (`META_MODE=recorded`); set `META_MODE=live` to call the real API with your own key.

Environment variables and secrets

Every variable is validated by a Zod schema at startup, so a missing or malformed value stops the app with a clear message instead of failing later. Values below are placeholders; real values live only in the dashboards and the team vault.

Variable	Example placeholder	Secret?	Set in
NEXT_PUBLIC_SUPABASE_URL	https://<project-ref>.supabase.co	No (public by design)	Vercel (preview, production), .env.local
NEXT_PUBLIC_SUPABASE_PUBLISHABLE_KEY	<publishable-key>	No (safe because RLS is on)	Same
SUPABASE_SERVICE_ROLE_KEY	<service-role-key>	Yes, critical	Vercel only; never `NEXT_PUBLIC_`
META_API_KEY	<meta-model-api-key>	Yes, critical	Vercel; local only if `META_MODE=live`

Variable	Example placeholder	Secret?	Set in
META_BASE_URL	https://api.meta.ai/v1	No	Vercel, .env.local (fallback switch to OpenRouter)
META_MODE	recorded or live	No	.env.local, CI
NEXT_PUBLIC_TURNSTILE_SITE_KEY	<site-key>	No	Vercel, .env.local (test key locally)
TURNSTILE_SECRET_KEY	<secret-key>	Yes	Vercel
AXIOM_TOKEN, AXIOM_DATASET	<ingest-only-token>, shelfvoice	Token yes	Vercel
BETTERSTACK_HEARTBEAT_URL	<heartbeat-url>	Yes	Vercel
DISCORD_ALERT_WEBHOOK_URL	<webhook-url>	Yes	Vercel
CRON_SECRET	<random 32+ characters>	Yes	Vercel
ADMIN_EMAILS	<comma-separated team emails>	No	Vercel
AI_DAILY_BUDGET_USD, AI_TOTAL_BUDGET_USD	15, 150	No	Vercel
RELEASE	set automatically from the git SHA	No	Build

Rules: `.env*` files except `.env.example` are git-ignored; GitHub secret scanning and push protection are on; a leaked key is rotated before anything else (document 03).

Database provisioning and migrations

Both hosted databases are created once on day 1 and from then on change only through migrations applied by CI or the documented command.

One-time provisioning (day 1):

1. Create Supabase projects `shelfvoice-staging` and `shelfvoice-prod` in region ap-south-1 (Mumbai) on the free plan.
2. In each: turn on "Enable RLS on new tables"; confirm `pg_trgm`, `pg_cron` and `pgcrypto` are available in Extensions.
3. Auth: enable Google provider with the OAuth client ID and secret; enable anonymous sign-ins (demo shop); set the site URL and redirect URLs to the matching Vercel domain.
4. Storage: buckets are created by migration, not by hand.

5. Link the CLI: `supabase link --project-ref <ref>` for each project (stored per developer, not committed).

Applying migrations:

Step	Command	Who	When
Local	`supabase db reset`	Developer	Every pull
Staging	`supabase db push --linked` from CI with the staging project ref	CI job	Merge to main, before the app deploys
Production	Same command with the production ref	CI job, requires the production environment approval in GitHub	Merge to main after staging succeeds
Types	`pnpm db:types`	Developer, in the same PR as the migration	Every schema change

Rules: migrations are additive (no dropping a column in the same release that stops using it); a migration that changes data runs inside a transaction; before any production migration that alters existing rows, take a manual backup (Backup and restore). Seed data (`seed.sql`) runs only locally and in staging; production gets the demo shop from `scripts/demo-shop.ts` .

CI/CD and deployment

GitHub Actions runs every check and does the production release in a fixed order (database first, then app), so a deploy can never run new code against an old schema.

Workflow	Trigger	Steps
ci.yml	Every PR and push	Install; lint; typecheck; Vitest with coverage gate; `supabase start` + `supabase test db` ; npm audit, Semgrep, secret and bundle checks; Playwright E2E with recorded AI and axe
preview (Vercel Git integration)	Every PR push	Vercel builds a preview against the staging database; URL posted on the PR

Workflow	Trigger	Steps
release.yml	Merge to main, after ci.yml passes	1. Migrate staging; 2. migrate production (GitHub environment `production`); 3. `vercel deploy --prod` with a Vercel token; 4. smoke test `/api/health/ready` and the demo storefront; 5. post the release SHA to Discord
nightly.yml	02:00 IST daily	Live promptfoo evals; production `pg_dump` backup artifact; results to Discord
perf.yml	Manual, before release	Lighthouse CI on storefront and dashboard; k6 against staging

Vercel's automatic production deploy from main is turned off (preview deploys stay on), so release.yml is the only path to production.

Release steps for a person, if CI is unavailable:

1. Confirm main is green locally: `pnpm lint && pnpm typecheck && pnpm test && supabase test db`.
2. `supabase db push --linked` against staging, then production.
3. `vercel deploy --prod` from main.
4. Open `/api/health/ready` and the demo storefront; post the SHA in the team channel.

During the freeze (R4), release.yml requires a manual approval on the production environment, so every production change is a deliberate decision.

Release checklist

Used for the submission release on Nov 4 and for any release during judging week; each line gets PASS, FAIL, BLOCKED, NOT RUN or NOT APPLICABLE in document 09.

- [ ] CI green on the release commit (all five gates in document 04)
- [ ] Nightly live eval run meets every threshold, or exceptions are written down
- [ ] Security pre-production checklist complete (document 03)
- [ ] Migrations applied to staging and production; types regenerated
- [ ] Production env vars present and validated (app starts; `/api/health/ready` is 200)
- [ ] Google consent screen in production mode; one real sign-in works on production
- [ ] Demo shop reset; "Try the demo shop" works in a private window

- [ ] Storefront, review page and QR checked on a real Android phone
- [ ] Uptime monitors, heartbeat and Discord alerts tested with one forced failure
- [ ] Credit used so far recorded; alerts at 50% and 80% configured
- [ ] Backup ran in the last 24 hours and one restore was rehearsed
- [ ] Lighthouse and k6 results within targets
- [ ] Demo video recorded from this exact release
- [ ] Previous release SHA noted for rollback

Rollback procedures

The app rolls back in under a minute; the database never rolls back by default, because migrations are additive and old code keeps working against the newer schema.

Situation	Action	Time
Bad app release (errors, broken flow)	In Vercel, promote the previous production deployment (Instant Rollback), then revert the commit on main	About 1 minute
AI behaving badly after a prompt change	Revert the prompt version in code and release; or set the AI breaker to OPEN from the admin page to pause AI features	1-5 minutes
Migration failed midway on production	Migrations run in a transaction, so nothing applied; fix and re-run	Minutes
Migration applied but data wrong	Write a forward-fix migration; if data was lost, restore the affected tables from the latest backup into a scratch schema and copy rows back	30-60 minutes
Meta key leaked or revoked	Rotate the key in Meta's dashboard, update Vercel env, redeploy	5-10 minutes
Region or access block on the Meta API	Switch `META_BASE_URL` to OpenRouter with an OpenRouter key; redeploy (credit does not transfer, so watch spend)	10 minutes

Triggers for rollback, decided by the on-call person without waiting for a vote: readiness check failing, 5xx above 5% for 10 minutes, sign-in broken, or a cross-shop data leak (which also follows the incident steps in document 03).

Backup and restore

The Supabase free plan has no automatic daily backups, so a nightly dump of production is our backup, and a restore is rehearsed once before submission.

Backup (nightly.yml, 02:00 IST):

```
supabase db dump --linked --data-only -f data.sql     # rows
supabase db dump --linked -f schema.sql               # schema (also in
migrations)
# both files encrypted with the team's backup passphrase, kept 7 days
# as a private GitHub Actions artifact
```

Restore drill (once, on staging, before the Nov 4 noon IST freeze):

1. Download the latest artifact and decrypt it.
2. `supabase db reset --linked` on staging (applies migrations to an empty database).
3. `psql "$STAGING_DB_URL" -f data.sql`.
4. Run `supabase test db` and open the demo storefront; record the time taken in document 09.

Storage files are not backed up: voice clips and photos expire within 30 days by design, and product cut-outs can be regenerated from shelf photos or re-uploaded.

Daily operations and judging-week routine

A 10-minute check each morning catches problems before a judge does.

Every morning (build week and judging week), on-call person:

- [ ] Overnight Discord alerts: none open, or each has an owner
- [ ] Better Stack: all monitors up; heartbeat from the daily jobs received
- [ ] Admin page: credit used vs $150, AI success rate, p95 latency
- [ ] Nightly eval results: thresholds still met
- [ ] Demo shop: sign in privately, run one voice command, open the storefront
- [ ] Backup artifact from last night exists

Judging week (Nov 5, 7:30 AM IST to the Nov 9 closing ceremony):

Rule	Detail
On-call rota	Three 8-hour shifts covering 24 hours (document 05); handover note in Discord at each change
Change policy	Only fixes for breakages; every production change needs a second teammate's approval
Demo shop	Reset nightly at 3 AM IST; also reset after any live demo
Supabase pause risk	Readiness check every 3 minutes keeps the project active
Credit	If spend passes 80%, raise the per-shop daily cap rather than cutting demo access

Troubleshooting

The most likely failures, how they show up, and the first thing to check.

Symptom	Likely cause	First check	Fix
Every AI call returns 401 or 403	Key wrong, revoked, or region blocked	`ai_calls.outcome` and Axiom for the status code	Rotate key; if region, switch `META_BASE_URL` to OpenRouter
Voice note gives empty transcript	Wrong audio format (not 16 kHz mono WAV) or silent clip	WAV header check log line; play the clip from Storage	Fix encoder; ask owner to record closer
Products land under the wrong item	Weak name matching	Proposal's transcript vs product aliases	Add aliases; the owner corrects on the confirm screen
Owner sees another shop's data	RLS policy missing or wrong (critical)	`supabase test db`; Security Advisor	Roll back now; incident steps in document 03
Storefront shows old prices	Cache not revalidated	Revalidation log line after catalog change	Trigger revalidation; check the tag name

Symptom	Likely cause	First check	Fix
Sign-in loops back to the sign-in page	Redirect URL mismatch	Supabase Auth URL settings vs Vercel domain	Add the exact domain to redirect URLs
Review submit returns 403	Turnstile token missing or wrong key pair	Browser console; Turnstile dashboard	Match site key and secret per environment
App fast locally, slow in production	Functions not in bom1, or cold database	Vercel function region; Supabase region	Pin region to bom1; keep-alive running
Supabase project paused	7 days without activity	Supabase dashboard status	Restore from dashboard; check the keep-alive check
Stock goes negative error	Sale logged larger than stock count	Product's movement log	Owner logs a correction first
Credit draining fast	Loop, retries or abuse of the public review page	`ai_spend_daily` by job; rate-limit hits	Lower caps; open the AI breaker; block the abusing IP

Dependency updates

Dependencies are pinned and changed on purpose, never by surprise in the week of a demo.

- Lockfile committed; AI SDK, Supabase and auth libraries pinned to exact versions.
- Dependabot opens weekly PRs for security updates; each goes through the full CI like any other PR.
- Critical or high vulnerabilities in production dependencies are fixed before release (document 03 triage rules); dev-only issues can wait with a note.
- From the R4 freeze until the Nov 9 closing ceremony, only security fixes are merged.
- Node stays on 24.x for the hackathon; Node 24 receives security fixes until Apr 2028.

Sources

Checked Oct 27, 2026.

- Vercel supported Node.js versions: 24.x default, 22.x available; set via `engines` in package.json

- Node.js release and end-of-life dates: Node 24 maintenance LTS until Apr 2028
- Vercel Hobby plan: one seat, non-commercial, 1-hour runtime logs
- Supabase CLI reference for agents: link, db push, db dump, test db commands
- Supabase Postgres 17 release notes